Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Autonomous Robotic Endoscope for Minimally Invasive Interventions			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio · Medical		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic devices & diagnostic devices	Interventional Therapeutic Devices	*Robotics/ Automation Machinery*	Robot Control and Intelligence Technologies
Concept of Proposed Technology	The proposed technology aims to develop an autonomous robotic endoscope for minimally invasive interventions that enables safe, precise, and task-aware operation in confined anatomical environments. The system integrates a highly dexterous continuum robotic endoscope, physics-grounded modeling, and vision-guided closed-loop control to support autonomous navigation, inspection, and target-focused manipulation under human supervision. By tightly coupling real-time endoscopic vision with robot state estimation and control, the proposed technology enables reliable autonomy beyond conventional teleoperated or assistive endoscopic systems. Key features:			

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	• Vision-guided perception and 3D scene understanding using monocular endoscopic imaging for real-time estimation of anatomical structures, camera pose, and spatial relationships • Physics-grounded continuum robot modeling and closed-loop control for accurate, stable, and safe motion of the robotic endoscope in constrained and deformable environments • Task-aware autonomous behaviors such as target tracking, view stabilization, surface scanning, and collision-aware navigation to support minimally invasive diagnostic and interventional procedures
Rationale for Joint R&D Project	The proposed project requires close collaboration between a research institute and an industrial partner to successfully translate autonomous robotic endoscope technologies into a practical system for minimally invasive interventions. A joint R&D framework is essential to integrate advanced perception, modeling, and autonomy algorithms with clinical-grade endoscopic robotic hardware, and to validate the proposed technology under realistic anatomical and operational conditions. Rationale for collaboration: • The research institute provides expertise in continuum robotic design, physics-grounded kinematic modeling, vision-guided perception, and autonomous control algorithms for robotic endoscopy • The Korean SME partner provides capabilities in endoscopic robot hardware development, system integration, regulatory-oriented design, and real-world experimental validation • Effective co-design of autonomy software and endoscopic robotic hardware is necessary to meet stringent safety, reliability, size, and cost constraints required for minimally invasive interventional applications Expected benefits of joint R&D: • Accelerated transition from laboratory-scale autonomy research to deployable robotic endoscope prototypes • Reduced technical risk through iterative hardware–software co-development, system integration, and experimental validation • Creation of a scalable technological foundation for commercialization of intelligent and autonomous robotic endoscopy systems
Key Words	1. Autonomous Robotic Endoscopy 2. Vision-Guided Closed-Loop Control 3. Continuum Robotic Systems 4. Endoscopic 3D Scene Reconstruction 5. Minimally Invasive Interventions

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop an autonomous robotic endoscope that can reliably operate in real-world minimally invasive interventional settings by integrating vision-based perception, physics-grounded modeling, and closed-loop control. The proposed system aims to enable safe, precise, and task-aware endoscopic navigation and inspection under realistic anatomical and operational conditions. The specific R&D objectives are: • To develop vision-guided perception and 3D scene understanding algorithms that enable robust estimation of anatomical structures, camera pose, and spatial relationships from monocular endoscopic imaging • To develop physics-grounded kinematic modeling and closed-loop control methods for accurate, stable, and safe motion of continuum robotic endoscopes in confined and deformable environments • To develop task-level autonomous endoscopic behaviors such as target tracking, view stabilization, surface scanning, and collision-aware navigation to support minimally invasive interventional procedures			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Endoscopic Navigation and Positioning Performance (95%>):** Achieve an autonomous endoscopic navigation and positioning success rate greater than 95% in confined and complex anatomical environments, including narrow lumens, curved pathways, and limited visibility, while maintaining collision-free operation under predefined safety constraints. **Vision-Based Perception and Scene Understanding Performance (85%>):** Achieve a vision-based perception and 3D scene understanding success rate greater than 85% using monocular endoscopic imaging, enabling reliable estimation of camera pose, anatomical structures, and spatial relationships under partial observability and challenging imaging conditions. **Autonomous Endoscopic Task Execution Performance (80%>):** Achieve an autonomous task execution success rate of at least 80% for representative endoscopic tasks, such as target tracking, view stabilization, and surface scanning, while maintaining stable, safe, and repeatable operation suitable for minimally invasive interventional workflows.			
R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from modeling and laboratory experiments to integrated system validation in realistic minimally invasive interventional settings. **Year 1: Modeling, Perception, and Core Autonomy Algorithm Development** • Develop a physics-grounded kinematic and dynamic modeling framework for continuum robotic endoscopes to enable accurate state estimation and motion prediction • Implement vision-based perception and 3D scene reconstruction algorithms using monocular endoscopic imaging for camera pose estimation and anatomical scene understanding			

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	• Develop core vision-guided control and autonomy modules, including view stabilization and target-aware motion primitives • Validate core models and algorithms through simulation and benchtop phantom experiments, focusing on accuracy, stability, and safety-related metrics Year 2: Laboratory-Scale System Integration and Closed-Loop Control Validation • Integrate perception, modeling, and control modules on a robotic endoscope hardware platform in a laboratory environment • Develop closed-loop visual servoing and safety-aware control strategies under realistic sensing noise and hardware constraints • Optimize autonomy algorithms for real-time performance, robustness, and repeatability on embedded and robotic control systems • Conduct repeated laboratory-scale experiments to evaluate navigation accuracy, task execution success rates, and system robustness Year 3: Integrated Autonomy Demonstration and Translational Preparation • Demonstrate task-level autonomous endoscopic behaviors, such as navigation in confined anatomical pathways, surface scanning, and target tracking, under human supervisory control • Perform system-level validation experiments using realistic anatomical models to verify safety, reliability, and performance consistency • Refine system architecture and control strategies based on experimental results to improve stability, usability, and robustness • Assess commercialization and industrial transfer potential, including scalability, manufacturability, and alignment with medical device development pathways

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the development of core autonomy algorithms and system-level intelligence for the proposed autonomous robotic endoscope technology. The institute will lead research on vision-based perception, physics-grounded modeling, decision-making, and closed-loop control methodologies required to achieve safe, precise, and reliable endoscopic operation in minimally invasive interventional settings. The specific roles and responsibilities of the research institute include: • Development of vision-guided perception, 3D scene understanding, and autonomous control algorithms for robotic endoscopy, including camera pose estimation, target tracking, and task-level autonomy • Development of physics-grounded kinematic modeling and closed-loop control methods for continuum robotic endoscopes to enable accurate, stable, and safe motion in confined anatomical environments • Design of the overall system architecture and autonomy integration framework, including interfaces between perception, modeling, control, and hardware modules • Simulation- and laboratory-based development, algorithm validation, and performance analysis of the proposed technologies • Provision of technical guidance and close collaboration with the industrial partner to support system integration, experimental evaluation, and technology transfer
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, integration, and validation of the robotic endoscope hardware and system-level implementation required for the proposed autonomous robotic endoscope technology. The SME partner will play a critical role in translating autonomy algorithms into a practical, reliable, and deployable endoscopic robotic system suitable for minimally invasive interventional applications. The specific roles and responsibilities of the Korean SME partner include: • Design and development of robotic endoscope hardware, including the continuum endoscope structure, actuation mechanisms, sensing components, and system-level hardware modules • System integration of perception, modeling, and autonomy algorithms with robotic hardware, sensors, and embedded control platforms • Execution of laboratory-scale and realistic experimental validation, including system testing, data collection, performance evaluation, and safety assessment • Optimization of system reliability, robustness, and manufacturability based on experimental results and hardware constraints • Contribution to commercialization and industrialization planning, leveraging industrial expertise in product development, regulatory-oriented design, and manufacturing capabilities

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**The proposed joint R&D project is expected to deliver a prototype autonomous robotic endoscope capable of safe, reliable, and task-aware operation in minimally invasive interventional environments.** The project will advance the state of the art in vision-guided perception, physics-grounded modeling, and closed-loop autonomous control for robotic endoscopy, achieving measurable improvements in navigation accuracy, system stability, and operational robustness. From an industrial perspective, this project will provide the Korean SME partner with **core autonomous endoscopic robotic technologies and system integration know-how applicable to next-generation robotic endoscopy platforms**. The outcomes will support technology commercialization and product development, enabling the development of scalable, reliable, and cost-effective robotic endoscope systems for clinical and interventional use. More broadly, the project is expected to **strengthen the global competitiveness of Korean medical robotics SMEs**, accelerate international technology collaboration, and contribute to the growth of the intelligent and autonomous medical robotics industry through successful translation of research outcomes into deployable products.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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